

Robotics and the VZ

By Ben Hobson and Dave Maunder.

Written November, 1991.

In the last few months we have been experimenting with computer controlled vehicles. Not much is required in the form of electronics. The major difficulty is the constructing of a suitable vehicle. One way we built a vehicle was out of Technic Lego using cogs to gear down the drive to tank like tread. This particular model originally was the mid 1980's Technic Lego Bulldozer 856. Another method is to convert an old remote controlled car. These can come in two types, one is controlled via a four way cable and the other is via its original radio controller. Either will do as long as you can change polarity to the motors with a double pole relay. In one of the prototypes, a radio controlled car was used. The transmitter was missing and as such the control circuitry was built into the actual car, with a twelve way cable running to the computer and power supply.

The circuit is a modification of a circuit which appeared in the USBORNE book, "Practical things to do with a microcomputer". The original circuit did not have the capacitors and resistors, as it was designed for a BBC or similar computer which obviously has the 5volt supply entirely separate from the data lines. As such it did not work on the VZ. The resistors simply provide a current difference sufficient to turn on the transistors. The diodes and capacitors absorb any back EMF spikes produced by the switching of the replays. Without them, fast switching will at certain times hang the VZ. The single pole relay is the power on/off relay, turning on / off power to the motors. This is necessary because the motor relays only reverse the polarity of the motors and such do not turn the motors off. In the circuit diagram the relays are represented by a little blob with a line either side. The blob is the common connection.

The bumper switches are two normally open micro switches. They are there to sense if your robot has hit anything. Connect them together via a strip of cardboard or plastic to form a semi circular shape. This way any object that is hit by the bumper will activate either or both switches. Two bumper circuits are shown. It is up to you which one to use if any. Two bumpers can be used, one at the front and one at the back.

Motors can be anything at all... Cassette player motors appropriately geared are ideal. Just make sure they rotate at the same speed or the vehicle will veer to one side. The power supply can be either battery or a plug pack transformer. Just don't try and use the VZ's +5v supply. If for some reason the motors stall, they will place an excessive load on the VZ's power supply and may damage it. Besides that it is doubtful if enough current can be supplied anyway. The VZ's regulator can only supply one amp and most of this used up by memory expansions, disk controllers etc.

To use the printer interface for the vehicle a small modification is necessary. A 5volt supply must be available at the Centronics plug. Open up your printer interface and position it so the edge connector is forward and the PCB tracks are uppermost. Solder a wire to pin 4 of the edge connector. This can be located by counting to the fourth pin from the left on the row of pins closest to the VZ side of the interface. Now wind the wire around the cable, being sure to loop it through the straining clamp and connect it to a spare pin on the Centronics plug. Centronics plugs have the pin numbers labeled to avoid confusion. In most cases PIN 36 is not used but you should check your printer manual to be sure. If pin 36 is used by the printer the 5v will disturb operation of the printer. However 99% of printers stick to the industrial standard layout and according to that PIN 36 is not used. Reassemble the interface and check that 5v is available between PIN 36 (or your choice of pin) and PIN 16. If all is well you can build your vehicle.

The Centronics pins and their functions that are used is as follows.

PIN 2 - D0	data line 0	PIN 11 – Busy/Ready
PIN 3 - D1	data line 1	PIN 16 - Ground
PIN 4 - D2	data line 2	PIN 36 - 5volt
PIN 5 - D3	data line 3	
PIN 6 - D4	data line 4	
PIN 7 - D5	data line 5	
PIN 8 - D6	data line 6	
PIN 9 - D7	data line 7	

Writing software is easily achieved from BASIC. A simple OUT statement turns the data lines on or off. To calculate these values place a binary 1 in the appropriate data line number that corresponds to the relay you want turned on and convert this to decimal. Then type something like OUT 14, 64 for the relay that is connected on Pin 8 - D6. The port you send it to may depend on your version of printer interface, but try 12, 13, 14, 15. The VZ uses ports 0 to 15 for the Printer Interface.

Port 14 was used successfully with a VZ200 printer interface – “late version” connected to a VZ200 & VZ300 computer.

OUT 14, 00 to turn off all relays.

OUT 14, 01 to operate relay #1 on Pin 2 - D0.

OUT 14, 02 to operate relay #2 on Pin 3 - D1.

OUT 14, 04 to operate relay #3 on Pin 4 - D2.

OUT 14, 08 to operate relay #4 on Pin 5 - D3.

OUT 14, 16 to operate relay #5 on Pin 6 - D4.

OUT 14, 32 to operate relay #6 on Pin 7 - D5.

OUT 14, 64 to operate relay #7 on Pin 8 - D6.

OUT 14, 128 to operate relay #8 on Pin 9 - D7.

OUT 14, 25 to operate relays #1, #4, #5. (25 decimal = 00011001 binary)

OUT 14, 204 to operate relays #3, #4, #7, #8 (204 decimal = 11001100 binary)

In the circuit provided, motor one is sitting on relay #1, motor two is controlled by relay #3, and on/off power to both motors is controlled by relay #2.

To test your bumper you can use a simple line such as :

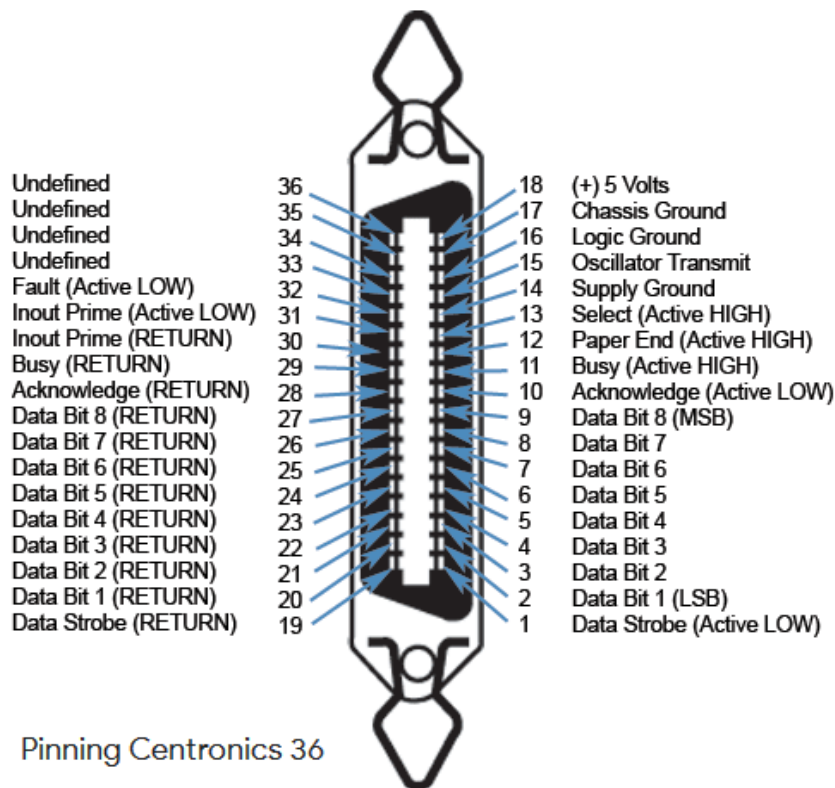
```
A=INP(0) : IF A = 24 THEN GOSUB 1000 ELSE GOTO 20
```

The value for A can be discovered by the simple program.

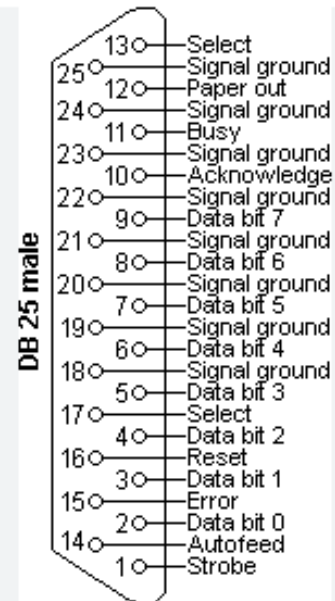
```
10 A=INP(0) : PRINT A : GOTO 10
```

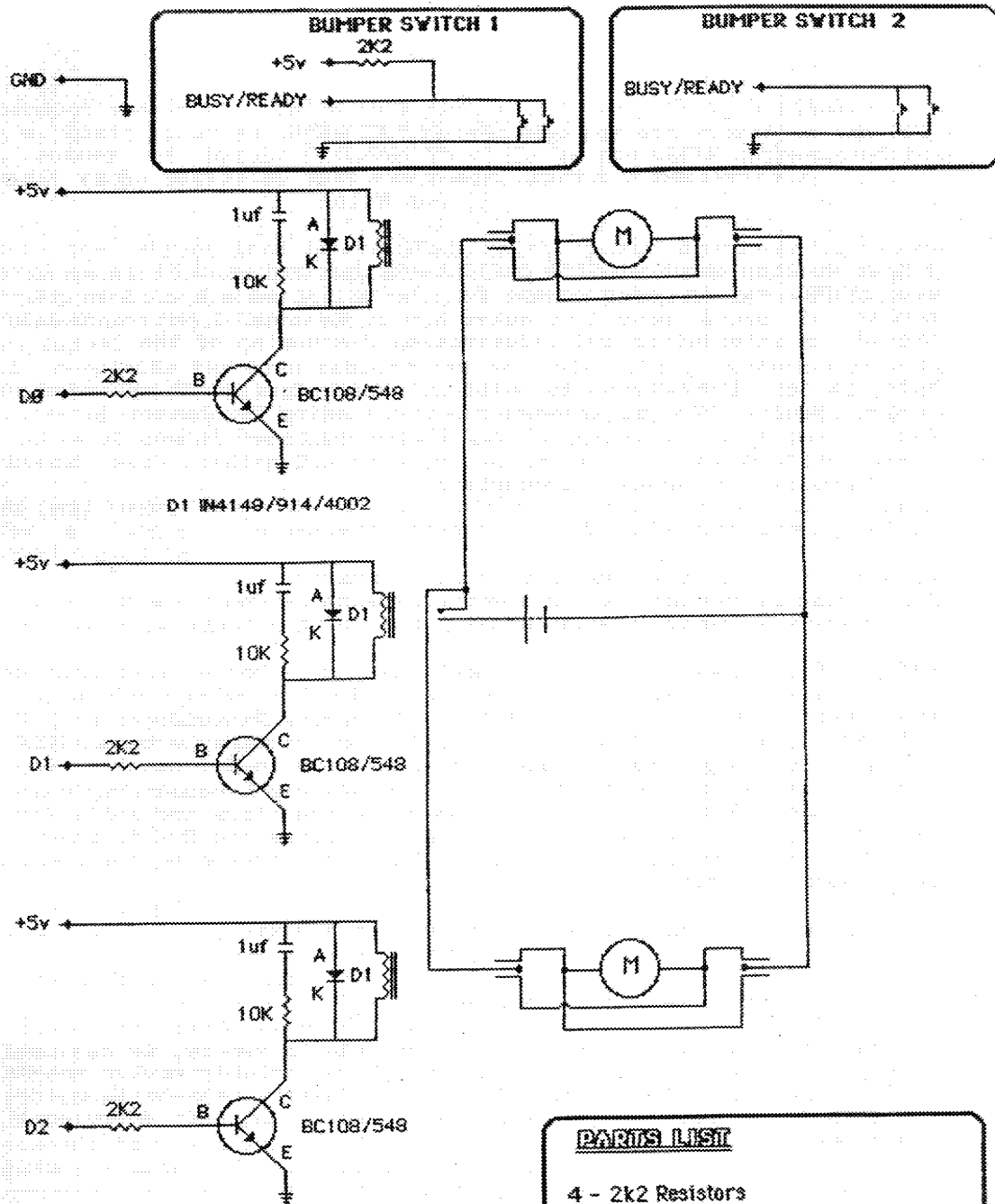
Then press the bumper switches a few times and observe what the resulting number is when the switch is open and closed.

Parallel Port pin-out included if the reader wishes to venture this project over onto the parallel port on the IBM compatible PC computer systems.



Parallel cable pinout

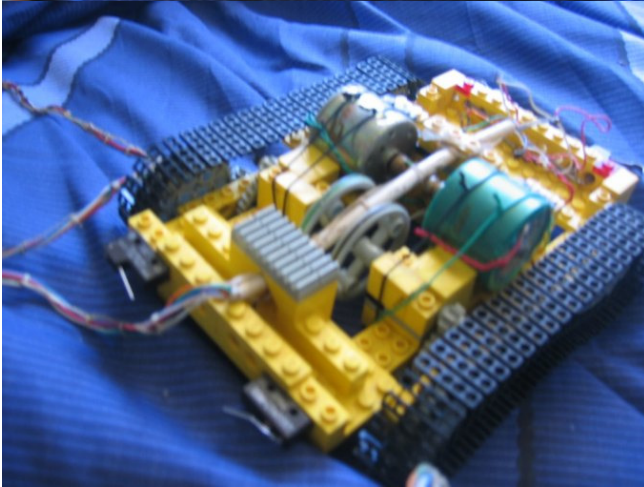
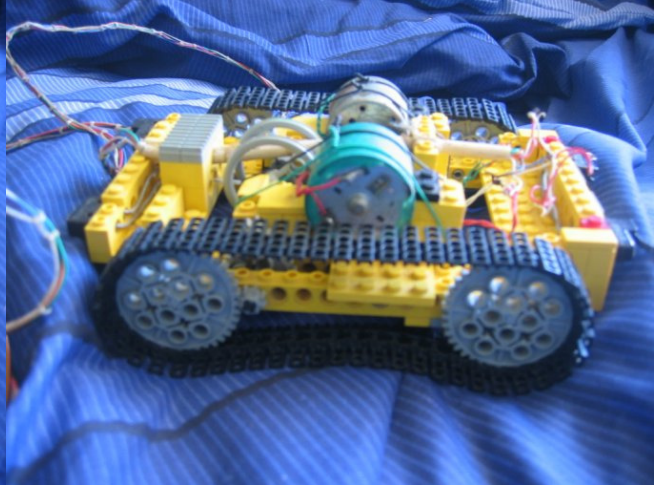
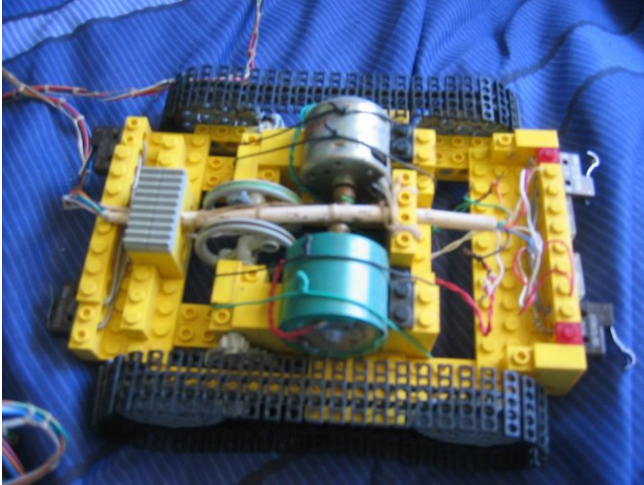
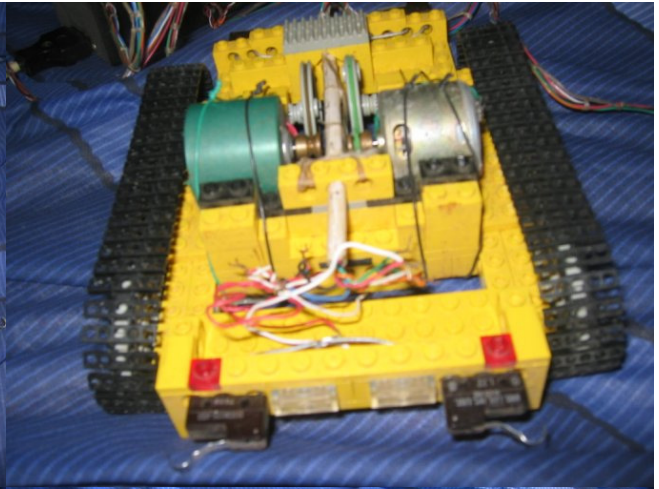
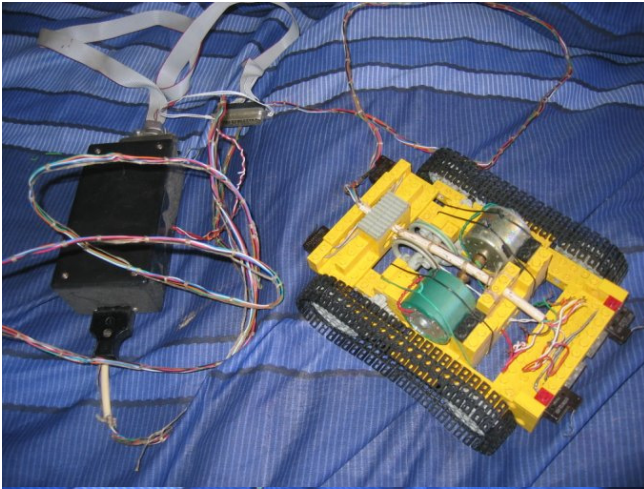




Computer Controlled Robot
 Drawn by Ben Hobson
 14-08-1991
 For use on Parallel Printer Port

PARTS LIST

- 4 - 2k2 Resistors
- 3 - 10k Resistors
- 3 - 1uF Capacitors
- 2 - Double Pole Double Throw 5v Relays
- 1 - Single Pole Single Throw 5v Relay
- 3 - BC108 or similar transistors
- 3 - IN4001 or similar Diodes
- 4 - N.O. Microswitches
- 2 - Motors
- 1 - Battery to Suit motors



http://www.youtube.com/watch?v=7LMxXY12E_I
<http://www.youtube.com/watch?v=s5dvoxj0Mn8>
<http://bushy556.50megs.com/vz/vz-robot.html>

What you need for the robot's control circuit

Piece of Veroboard about 8cm x 8cm (with holes 0.1 inches apart).

2 double-pole changeover subminiature relays, coil voltage 6V d.c., coil resistance 250 ohms (or anything over 150 ohms will do).

1 single-pole subminiature relay, coil voltage 6V d.c., coil resistance 250 ohms (or anything over 150 ohms).

3 transistors type BC107 or BC108

3 x 2.2K resistors

3 diodes type IN4001, IN4002 or IN4003. (Do not use Zener diodes.)

Hook-up wire

Sticky tape e.g. masking tape

Battery to match your motors, e.g. a 9V battery for a 9V motor.

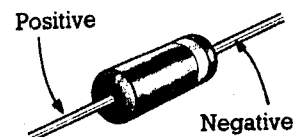
Plug and lead for user port.

Soldering the components on to the Veroboard

Label the holes and tracks on the Veroboard as shown on page 37. Make sure the ends of the tracks are labelled with letters and the holes are labelled down the side with numbers. There is a key over the page telling you in which holes to put the components. It does not matter which way round you solder resistors into the Veroboard but it is very important that you put the legs of diodes, transistors and relays in the right way round. * You can find out how to identify each leg on a transistor on page 36.

Diodes

A diode only allows the current to flow through it in one direction. The end with the stripe marks the negative leg and this goes towards the 0 volts, or negative, side of the circuit.

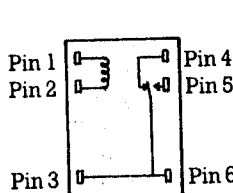


Relays

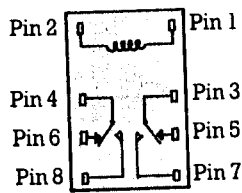
A relay is a tiny electronic switch. A double-pole relay has two switches inside and a single-pole has one. The double-pole relays in the robot's circuit switch the

motors on and make them go either forwards or backwards. Each motor drives a wheel. The single-pole relay turns the power supply on and off.

Circuit diagram for single-pole relay



Circuit diagram for double-pole relay



Place your relays over these diagrams.



Single-pole



Double-pole

When you buy your relays, ask for circuit diagrams showing how they work inside and compare them to these. If a diagram is labelled "pin view", you hold the relay with its legs facing you to look at it. Otherwise, you look at the relay from the top. If the circuit looks the same but the pins are labelled differently, substitute the numbers used here.

The relays you use must have their pins in the same position and the same distance apart, as the ones shown here or they will not fit into the right holes on the Veroboard. To make sure you buy suitable relays, take this book with you to the shop and place the relays over the diagrams above.

*There are instructions for how to solder on page 44.

Assembling the robot

These pages show how to make the robot's control circuit and how to connect it to the computer. Follow the instructions very carefully, as a single error will stop the circuit working.

- 1.** Solder the legs of the components into these holes on the Veroboard.

Double-pole relay:

Pin 1 - Z5
Pin 2 - Z8
Pin 3 - W5
Pin 4 - W8
Pin 5 - U5
Pin 6 - U8
Pin 7 - S5
Pin 8 - S8

Double-pole relay:

Pin 1 - P5
Pin 2 - P8
Pin 3 - M5
Pin 4 - M8
Pin 5 - K5
Pin 6 - K8
Pin 7 - I5
Pin 8 - I8

Single-pole relay:

Pin 1 - A8
Pin 2 - B8
Pin 3 - F8
Pin 4 - A5
Pin 5 - B5
Pin 6 - F5

Diode:

Positive - B10
Negative - A10

Diode:

Positive - P10
Negative - P3

Diode:

Negative - Z3
Positive - Z10

***Transistor:**

Collector - B13
Base - C13
Emitter - D13

Transistor:

Collector - P12
Base - O12
Emitter - N12

Transistor:

Collector - Z12
Base - Y12
Emitter - X12

Resistor: C15 and C19

Resistor: O14 and O19

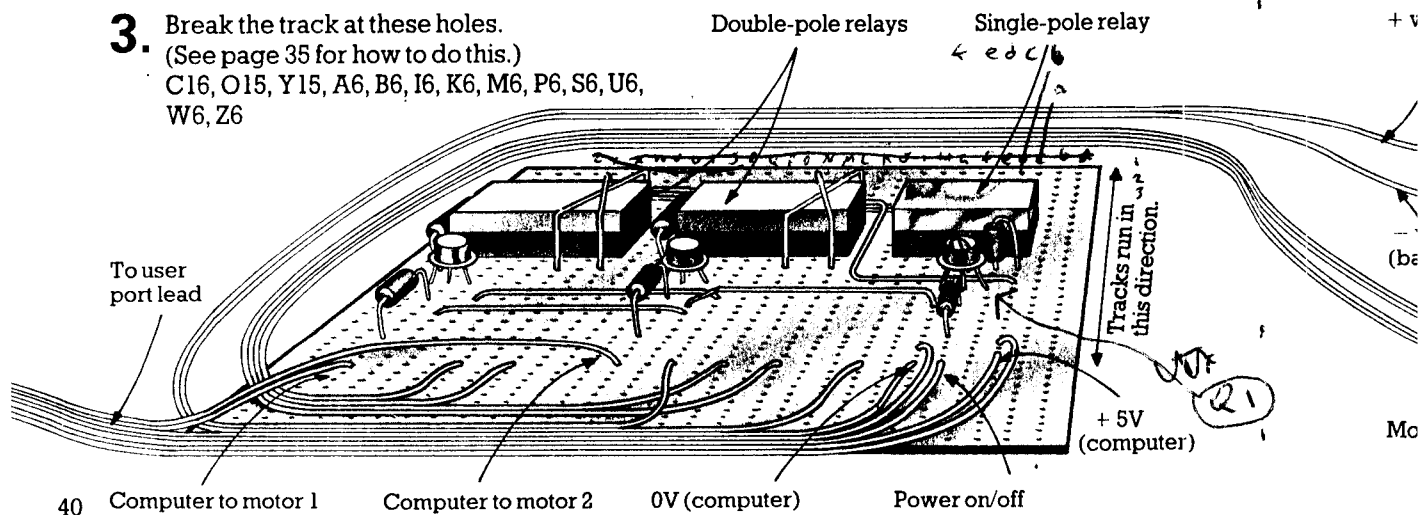
Resistor: Y14 and Y19

- 2.** Cut 11 pieces of wire each about 10cm long and strip about 1cm of the plastic coating from each end using wire strippers. Connect the following pairs of holes with the pieces of wire, soldering each end to the Veroboard underneath.

D15 and N15
N16 and X16
M14 and W14
K3 and I10
S3 and U10
I3 and K10

U3 and S10
A12 and P1
Z2 and P2
W2 and M2
M3 and B3

- 3.** Break the track at these holes.
(See page 35 for how to do this.)
C16, O15, Y15, A6, B6, I6, K6, M6, P6, S6, U6, W6, Z6



*See page 36 for how to identify each leg.

Connecting up the robot's circuit

1. Cut seven pieces of wire about 3m long. Strip the plastic from each end. Solder one end of each wire into the following holes, labelling each wire with a sticky label or piece of masking tape.

Wire 1 - A23	+5V (computer)
Wire 2 - C23	Power on/off
Wire 3 - D23	0V (computer)
Wire 4 - F23	+ volts (battery)
Wire 5 - Y23	Computer to motor 1
Wire 6 - M23	- volts (battery)
Wire 7 - O23	Computer to motor 2

2. Cut four pieces of wire about 20cm long. Strip each end and solder one end of each into the following holes, labelling the wires as you go.

Wire 8 - S23	Motor 1A
Wire 9 - U23	Motor 1B
Wire 10 - K23	Motor 2A
Wire 11 - I23	Motor 2B

3. Identify the wires in the user port's lead which connect to output pins 0, 1 and 2 in the user port. (You may need to ask the supplier when you buy the lead.) Strip 1cm of plastic from the ends of these wires and solder them to the labelled wires on the robot's circuit as follows:

Label on wire	User port lead
Computer to motor 1	Wire to output pin 0
Computer to motor 2	Wire to output pin 1
Power on/off	Wire to output pin 2
+5V (computer)	Wire to 5V pin
0V (computer)	Wire to 0V pin

4. Tape the following two wires to the ends of the battery:

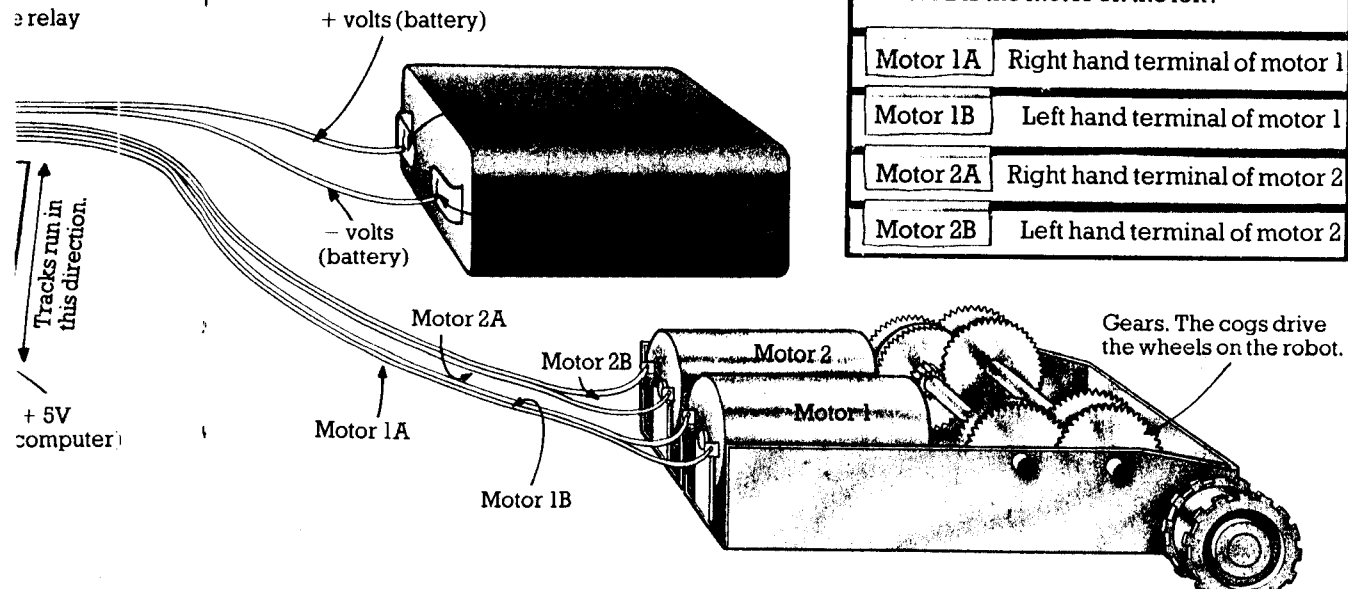
Label on wire	Battery
+ volts (battery)	+ volts terminal
- volts (battery)	- volts terminal

The last four wires connect to the motors. Motor 1 is the motor on the right as you look down at the robot from above and Motor 2 is the motor on the left.

Motor 1A	Right hand terminal of motor 1
Motor 1B	Left hand terminal of motor 1
Motor 2A	Right hand terminal of motor 2
Motor 2B	Left hand terminal of motor 2

working.

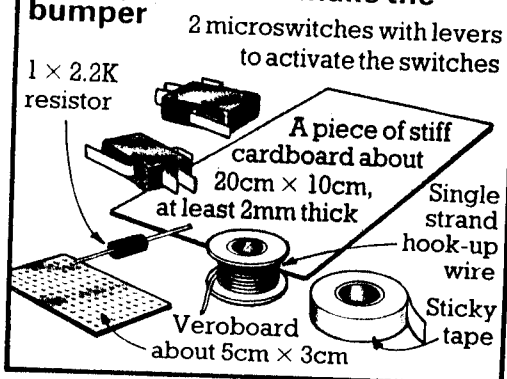
a relay



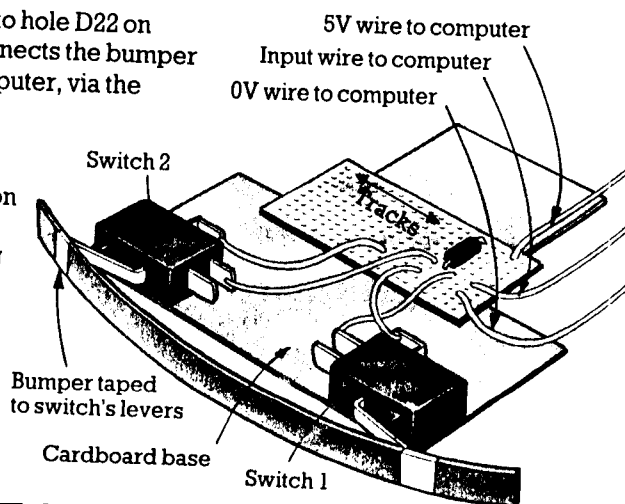
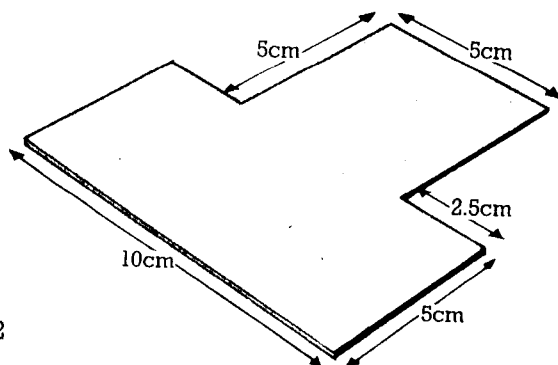
How to make the robot's bumper

The robot has a bumper attached to its front, with a switch at either side. The switches are activated when the robot bumps into something and the computer's program tells the robot to reverse and change direction.

What you need to make the bumper



1. Cut pieces of wire to the following lengths and strip 1cm of the plastic from either end. Solder one end of each wire into these holes on the Veroboard:
 2 pieces of wire about 25cm long – 2J
 – 2C
 2 pieces of wire about 5cm long – 7C
 – 11C
 1 piece of wire about 3cm long – 2E
 2 pieces of wire about 5cm long – 5E
 – 9E
 Solder the resistor into holes 6E and 6J
2. Solder the other end of the wire in 7C to the common (com or 1) connection on switch 1 and the other end of the wire in 11C to the com connection on switch 2. Solder the other end of the wire in 5E to normally open (NO or 3) on switch 1 and the other end of the wire in 9E to NO on switch 2.
3. Solder the end of the input wire in hole 2E to the wire in the user port lead which connects to input pin 3.
4. Solder the end of the 5V wire (in hole 2C) into hole A21 on the robot's control circuit board. This carries a 5V power supply to the bumper circuit from the computer, via the robot's control circuit.
5. Solder the end of the 0V wire in hole 2J into hole D22 on the robot's control circuit board. This connects the bumper circuit to the 0V power supply in the computer, via the robot's control circuit.
6. Cut a piece of cardboard this size for the base. Stick the circuit and both switches on to the base with tape. You can adjust the distance between the switches according to the width of your robot's body.



7. Cut a strip of cardboard about 18cm x 1.5cm for the bumper (the length will depend on the size of the robot). Tape it to the levers on the switches. Attach the bumper unit to the robot by taping it underneath or on the top at the front – wherever you can fix it firmly.

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Signal:
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F=7 (bina
User port

1	Pin 0	--
1	Pin 1	--
1	Pin 2	--
0	Pin 3	--
0	Pin 4	--
0	Pin 5	--
0	Pin 6	--
0	Pin 7	--

Here are
ways. Ins
is a 1 in th
the numb
(Address
swap the
you will n

10 (Set I
20 LET
30 POKE
40 LET
50 IF
60 GOTO
70 POKE
80 FOR
90 GOSL
100 NEXT
110 POKE
120 LET
130 IF R
140 POKE
150 FOR
160 GOSU
170 NEXT
180 GOTO
190 FOR
200 NEXT
210 RETUF

*Set DDR if n
pins are not c

A program for the robot

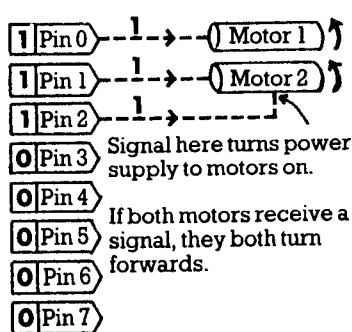
This program makes the robot move forwards until it hits something, and then reverse and swivel right or left before setting off again. Try building an obstacle course from piles of books and other objects and see if the robot can get round it. Before you can run the program, you need to insert some numbers which will make the program work.

Finding the correct numbers

Signals from the pins in the user port activate the motors. You send the signals by using the command POKE, followed by different numbers which make the robot go in different directions. The letters F, B, R, L and S in the program stand for the numbers which make the robot go forwards, backwards, right, left and stop. You need to experiment to find the right numbers for your computer and motors.

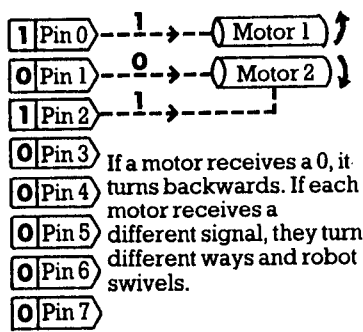
F=7 (binary 00000111)

User port Signal



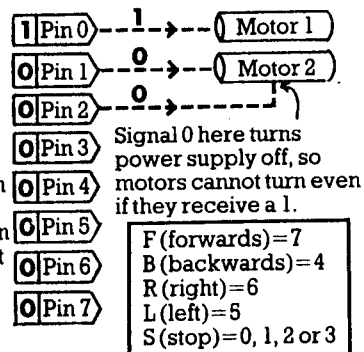
L=5 (binary 00000101)

User port Signal



S=1 (binary 00000001)

User port Signal



Here are three examples showing how different numbers make the robot move different ways. Inside the computer the numbers are converted to binary numbers and where there is a 1 in the number, a signal is sent out of the user port from the pin in the same position. Try the numbers shown on the right with your computer by using the command POKE (Address), number.* If they make the robot go in different directions from those listed, swap the numbers around. If you have connected the motors to pins other than 0, 1 and 2 you will need to try different numbers in the range 0 to 255.

```

10 (Set DDR if necessary)
20 LET Y=(Address)
30 POKE Y,F
40 LET Z=PEEK Y AND 8
50 IF Z<>0 THEN GOTO 70
60 GOTO 40
70 POKE Y,B
80 FOR I=1 TO 10
90 GOSUB 190
100 NEXT I
110 POKE Y,S
120 LET P=R
130 IF RND(1)>0.5 THEN LET P=L
140 POKE Y,P
150 FOR I=1 TO INT(RND(1)*20+10)
160 GOSUB 190
170 NEXT I
180 GOTO 30
190 FOR T=1 TO 100
200 NEXT T
210 RETURN
    
```

Robot moves forwards. (Substitute number for F.)

AND 8 makes computer look at signal on pin 3 in user port, which is connected to wire from bumper circuit.

Waits until switch is pressed (i.e. robot bumps into something) before going to line 70.

Robot goes into reverse. (Substitute number for B.)

Robot reverses while computer goes through delay loops, then stops.

Robot swivels right or left depending on whether random number is greater or less than 0.5. (Substitute numbers for R and L.)

Robot turns through random number of degrees.

Goes back to line 30 to start moving forwards in new direction.

*Set DDR if necessary (see your manual). Set pins 0, 1 and 2 to output, and pin 3 to input. The other pins are not connected to the robot so they do not matter.